

Dynamic Programming

→ 01 Knapsack
→ LCS

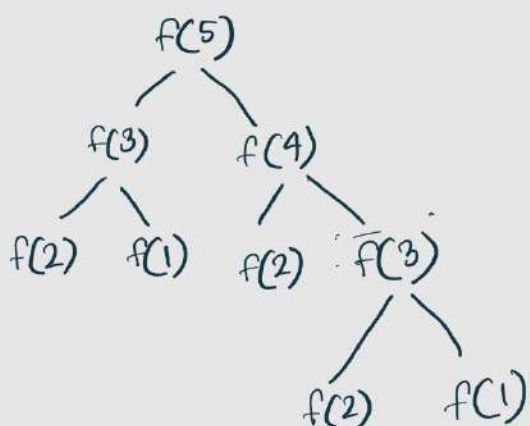
Overlapping subproblem/
optimal substructure

1, 1, 2

1, 1, 2, 3, 5

1 2 3 4 5

In Fibonacci series

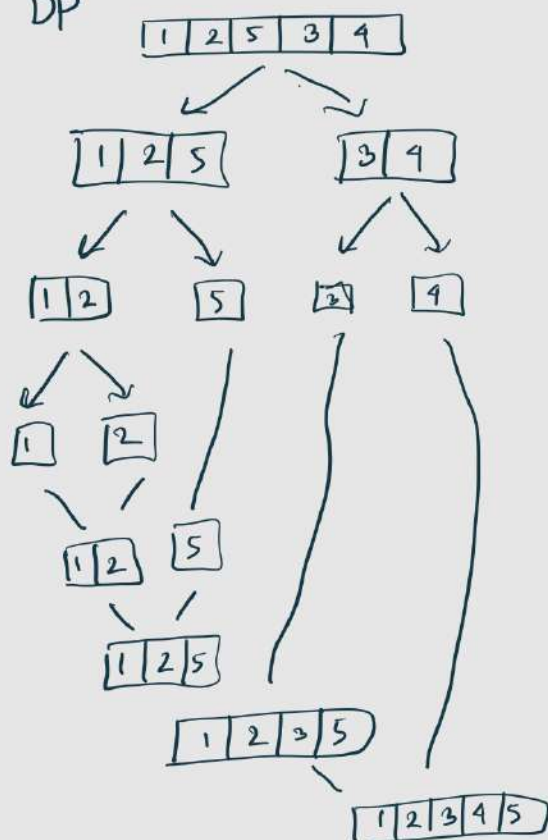


Memoization

arr[] = []

$O(NM) \rightarrow O(\min(N, M)) \rightarrow$ previous row store

Merge sort isn't a part of DP



LCS

[Longest Common Subsequence]

- X = ABCG (a) How many subsequence possible? ✓
Y = BDCAG (b) Find the LCS length ✓
(c) Find the LCS string ✓
(d) What does $dp[3][2] = 1$ mean? ✓
(e) Recursive Time Complexity ✓
(f) Iterative Time Complexity ✓
(g) How to make it better? ✓

(a) $|X| = 4$; Subsequence = 2^4
 $|Y| = 5$; Subsequence = 2^5

(b) $(n+1) \times (m+1)$ Landscape

X \ Y	E	B	D	C	A	G
E	0	0	0	0	0	0
A	0	0	0	0	1	1
B	0	1	1	1	1	1
C	0	1	1	2	2	2
G	0	1	1	2	2	3

Ans: 3

Code :

$$\begin{cases} 0 & i=0, j=0 \\ dp[i-1][j-1] + 1 & x_i = y_j \\ \max(dp[i-1][j], dp[i][j-1]) & \text{otherwise } [x_i \neq y_j] \end{cases}$$

(c) B C G

(d)

(e) $O(2^{n+m}) \rightarrow \text{Exponential}$

(f) $O(NM)$

(g) Previous row store

$$O(\min(N, M))$$

3. Today Alice has learned how to find the Longest Common Subsequence (LCS) of two given strings. Now she wants to find the LCS of “axyb” and “abyxb”. After hours of hard work, she has made this LCS table, M.

	empty	a	b	y	x	b
empty	0	0	0	0	0	0
a	0	1	1	1	1	1
x	0	1	1	1	2	2
y	0	1	1	2	2	2
b	0	1	2	2	2	3

- a.

CO1

Explain what you understand by the value $M[3][4] = 2$.

2
- b.

CO2

Find out the LCS String from the table. Show the steps of your work.

3

- 4 a.

CO1

String1: <Your ID> 24301121
String2: 13018828
- i.

ii.
- Create String2 in a way that the following constraints match. Write the string.

Write all three LCS strings (just write, no need to show any work).
- Length of String2 = 8 (eight) ✓

Length of $LCS(String1, String2) = 5$ (five) ✓

There are exactly three distinct LCS (of length = 5)

$$seq(i, j) = \max \begin{cases} seq(i, j - 1) - 1, \\ seq(i - 1, j) + p(i, j), \\ seq(i - 1, j) - 1 \end{cases}$$

where,

$$\begin{aligned} seq(i, j) &= 0, && \text{if } i = 0 \text{ or } j = 0 \\ p(i, j) &= 1, && \text{if } DNA1[i] = DNA2[j] \\ p(i, j) &= -2, && \text{otherwise} \end{aligned}$$

As you can see, the formula takes two indices, i and j of two DNA sequences given as input. Then it produces a similarity score as output. We want to apply this formula to compare the following DNA sequences.

index	1	2	3	4	5
DNA1	G	C	G	T	A
DNA2	C	T	G	A	G

Now answer the following questions:

- a)

CO1

Put appropriate values to fill the gaps in the line below (answer in your script):
 $seq(_, _)$ represents the final similarity score of the given DNAs.

02
- b)

CO1

Apply the formula to calculate a similarity score between the DNA sequences given above. Show your work with either a recursion tree or a memory table.

05
- c)

CO2

State the time complexity of this algorithm with proper reasoning.
You can assume that, the lengths of the DNA sequences are N and M respectively.

02
- d)

CO3

Mriaslun, a talented algorithmist, claims that he can implement the algorithm in an optimized way so that it only takes $O(\min(N, M))$ space. Can you do it too? Explain your idea.

01

- 4

Answer the following questions:
- a.

CO2

Can merge sort be written as a dynamic programming algorithm? Explain your answer.

02
- b.

We are calculating the longest common subsequence (LCS) between two strings,
 $S = X_1X_2X_3X_4$ and $T = Y_1Y_2Y_3Y_4$. To do this, we fill an array C where $C_{i,j}$ represents the length of the LCS between the prefix of length i from S and the prefix of length j from T.

Some of the entries in the array C are currently masked using different symbols other than integers as shown below.

	Y_0	Y_1	Y_2	Y_3	Y_4
X_0	0	0	0	0	0
X_1	0	0	0	0	1
X_2	0	0	0	1	1
X_3	0	0	1	1	♡
X_4	0	1	♣	♠	◇

$X_1 = Y_4$
 $X_2 = Y_3$
 $X_3 = Y_2$

CO1

i.

ii.

iii.

iv.

v.

vi.

CO3

vii.

What does the cell $C_{0,0}$ represent?

What can be said about the characters X_1 and Y_4 ?

What can be said about the characters X_2 and Y_4 ?

What is the value of ♡? 1

What is the value of ♣? 1

When we are filling up the i -th row of our dynamic programming table C, what rows do we need to have access to?

Given the observation above can we optimize our space complexity further?

01

01

01

01

01

01

02

0/1 Knapsack

$W = 8$

$W_i = \{3, 4, 5, 6\}$

$P_i = \{2, 3, 4, 1\}$

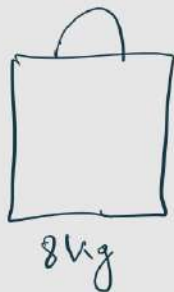
$n = 4$

Time complexity - Exponential

Recursive - $O(2^n)$

Dynamic - $O(nW)$

$n+1 : W+1$



Item	Weight (Profit)
1	3kg (2)
2	4kg (3)
3	5kg (4)
4	6kg (1)

wt \ W_i	0	1	2	3	4	5	6	7	8
0	0	0	0	0	0	0	0	0	0
2 (3kg)	0	0	0	2	2	2	2	2	2
3 (4kg)	0	0	0	2	3	3	3	5	5
4 (5kg)	0	0	0	2	3	4	4	5	6
6 (6kg)	0	0	0	2	3	4	4	5	6

Max Profit = 6
Item 3 & 1

$$M[i-1][wt] \quad \text{if } w_i > wt$$
$$\max(M[i-1][wt], M[i-1][wt - w_i] + v[i]) \quad \text{otherwise / } w_i \leq wt$$

3

Wye is a CSE BracU Student. The pre-advising week is approaching, and he is trying to find the most suitable course combination for next semester. As per his previous academic results, his credit limit **has been set to 9**. Different courses have different credits. He came up with values for these courses by consulting with students who have already taken them. He wants to select the most valued courses.

Courses, credits and values are provided below:

Course	PHY112	CSE330	CSE331	CSE370	CSE499
Credit	1	2	3	4	5
Value	10	8	16	13	17

CO1 a. Wye wants to try all possible combinations to find the best one. Help him by simulating a suitable **dynamic programming** approach. **Show** your work in detail, then **write** down the selected courses.

04

CO1 b. Zed, a friend of Wye, adopted a slightly different approach and considered the courses based on their per-credit value. **Determine** whether Zed could select more valued courses than Wye. Note that, selecting a course still means taking the whole with full credits.

02

CO1 c. If Zed is successful, he will send a message **Successful**; otherwise, **Unsuccessful**. Instead of fixed-length coding, he is willing to send the message using Huffman Coding. **Show** simulation of Huffman coding to encode his message.

04

→ Greedy

(a)

So, we have to do 0-1 knapsack to solve this problem:

So, the maximum value will be 43 and the courses taken are PHY112, CSE331 and CSE499.

Credit	0	1	2	3	4	5	6	7	8	9
Item	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	10	10	10	10	10	10	10	10	10
2	0	10	10	18	18	18	18	18	18	18
3	0	10	10	18	26	26	34	34	34	34
4	0	10	10	18	26	26	34	34	39	39
5	0	10	10	18	26	26	34	39	39	43

(b)

Zed took the courses on a Greedy approach.

Course	Value/Credit
PHY 112	10
CSE330	4
CSE331	5.33
CSE370	3.25
CSE499	3.4

So, he takes PHY112 (1 credit) → 10 value

CSE331 (3 credit) → 15.99

CSE330 (2 credit) → 8

6 credit → 34

Zed will get less value than Wye.

→ 34

Theme:

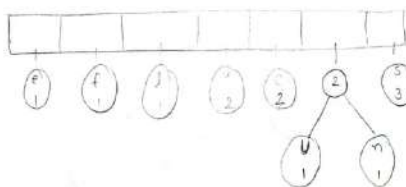
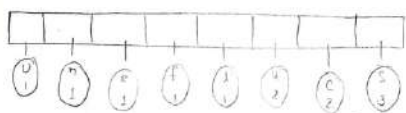
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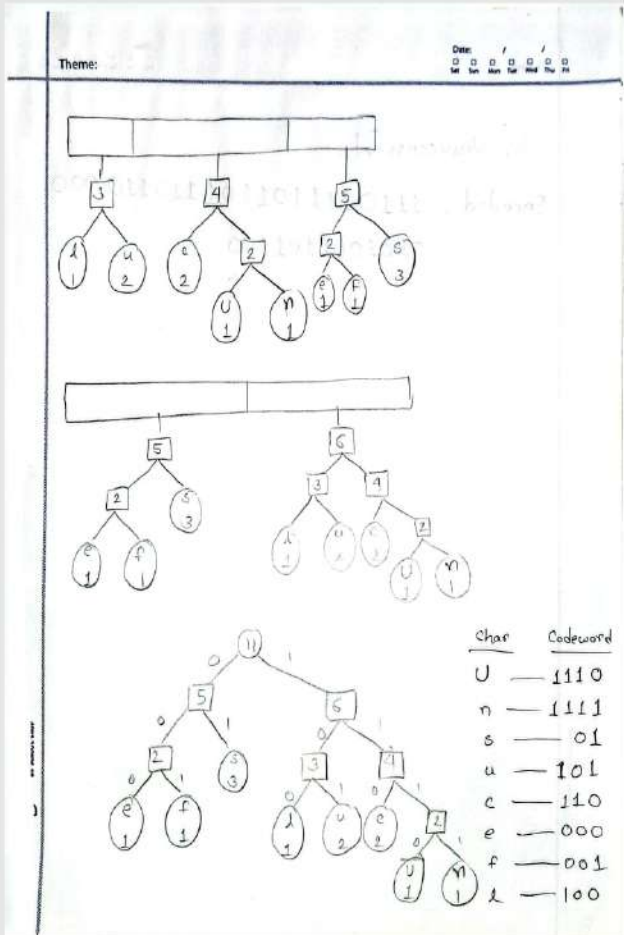
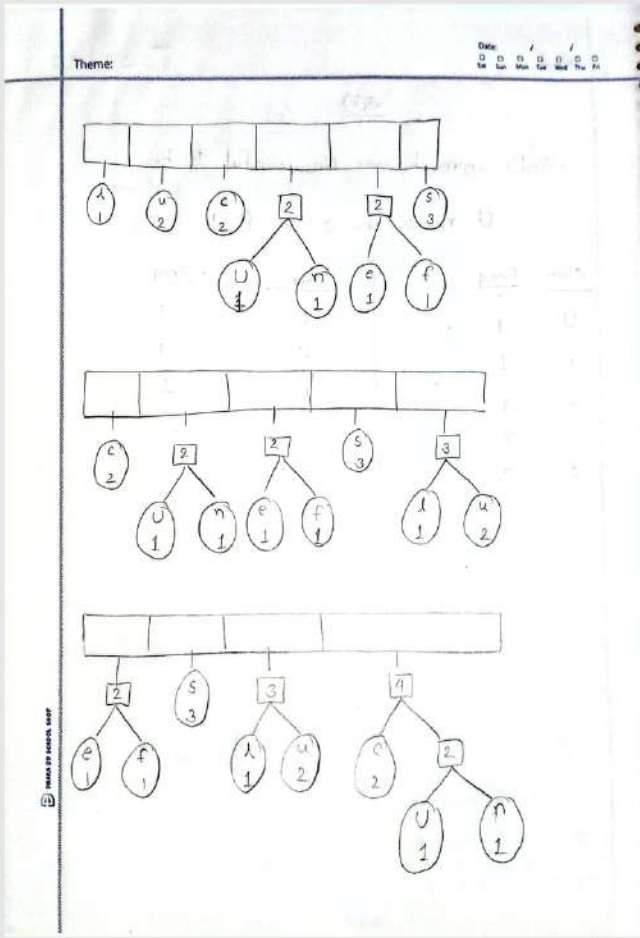
3(c)

Zed's approach was Unsuccessful. It has:

U n s u c e f l

Char	Freq	Char	Freq
U	1 ✓	e	1 ✓
n	1 ✓	p	1 ✓
s	3	l	1 ✓
u	2 ✓		
c	2 ✓		





Theme: _____ Date: / /

Sat Sun Mon Tue Wed Thu Fri

So, Unsuccessful.

Encoded : 1110111101101110110000
0101001101100

